

Norway

SLEIPNER

Volve F

F-10

F-10 A

Plan: F-10 A rev#7 LHULT 171115

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10 A	Survey Calculation Method:	Minimum Curvature
Design:	F-10 A rev#7 LHULT 171115	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-10			
Well Position	+N/-S	-3.91 m	Northing:	6,478,559.62 m
	+E/-W	2.23 m	Easting:	435,052.25 m
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-10 A				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	04.09.2015	-1.01	71.64	50,516

Design	F-10 A rev#7 LHULT 171115			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	1,388.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-3.91	2.23	25.52

Survey Tool Program	Date	01.02.2016		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
150.00	1,388.90	8 1/2" Keeper Gyro (F-10)	Keeper, cont	Gyro Tool from SDC
1,388.90	2,975.27	F-10 A rev#7 LHULT 171115 (F-10 A)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio	
2,975.27	3,969.68	F-10 A rev#7 LHULT 171115 (F-10 A)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio	

Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,388.90	27.61	119.42	1,330.36	-162.62	271.37	-27.26	0.000	0.000	0.000
1,398.90	27.61	119.42	1,339.22	-164.90	275.41	-27.58	0.000	0.000	0.000
1,418.90	26.46	116.77	1,357.03	-169.18	283.42	-27.99	2.500	-1.731	-3.969
1,827.57	14.42	298.79	1,754.88	-186.42	321.76	-27.03	3.000	-0.884	-13.065
2,339.82	14.42	298.79	2,251.00	-124.99	210.00	-19.75	0.000	0.000	0.000
2,396.29	17.36	290.16	2,305.31	-118.70	195.93	-20.14	2.000	1.562	-4.588
2,620.66	17.36	290.16	2,519.47	-95.64	133.10	-26.39	0.000	0.000	0.000
3,554.99	91.52	26.81	3,084.00	509.65	237.06	564.63	3.000	2.381	3.103
3,969.68	91.52	26.81	3,073.00	879.63	424.05	979.07	0.000	0.000	0.000

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Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
- hit/miss target									
- Shape									
T4 NW	0.00	360.00	3,073.00	879.78	424.13	6,479,443.00	435,474.00	58° 26' 58.4626 N	1° 53' 40.1755 E
- plan misses target center by 0.17m at 3969.68m MD (3073.00 TVD, 879.63 N, 424.05 E)									
- Rectangle (sides W100.00 H15.00 D30.00)									
T7 NW	0.00	360.00	3,084.00	509.65	237.06	6,479,073.00	435,287.00	58° 26' 46.4026 N	1° 53' 29.0205 E
- plan hits target center									
- Point									

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
201.70	201.70	30"	30.000	36.000	
1,388.90	1,330.36	20" Surface Casing	20.000	26.000	
2,812.33	2,700.00	9 7/8" Production Casing	9.875	12.250	
3,968.00	3,073.04	7" Production Liner	7.000	8.500	

Checked By: _____	Approved By: _____	Date: _____
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